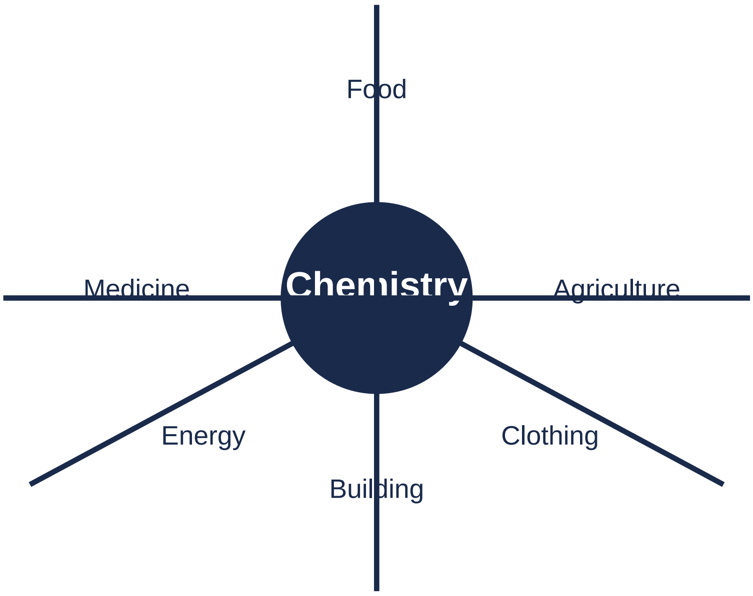
Unit 1: Chemistry and Its Importance

Complete Study Notes – Grade 9 Chemistry



1.1 Definition and Scope of Chemistry

Chemistry is the science that deals with the **properties**, **composition**, and **structure** of substances (elements and compounds), the **transformations** they undergo, and the **energy** that is released or absorbed during these processes.

A **substance** is a particular kind of matter with uniform properties. For example, gold, silver, water, soap, and table salt are all substances. Every substance has its own unique **properties** – attributes, qualities, or characteristics – by which we can distinguish it from other substances.

Composition refers to the nature of something's ingredients or constituents – the way in which a whole or mixture is made up. For instance, table salt (sodium chloride) is chemically composed of the elements sodium and chlorine. Stainless■steel spoons are solid solutions (alloys) of chromium, carbon, and other elements.

Structure is the arrangement of and relationships between the parts or elements of something complex. For example, a school building has a roof, ceiling, doors, windows, walls, and floor arranged in a certain order; this arrangement is the structure of the building.

Transformation is a marked change in form, nature, or appearance. Every substance in our environment is continuously changing due to external and internal forces. These transformations are accompanied by **energy changes** – energy is either absorbed or released during a chemical transformation.

The Scope of Chemistry – Five Main Disciplines

Physical Chemistry: The study of macroscopic properties, atomic properties, and phenomena in chemical systems. Physical chemists study rates of chemical reactions, energy transfers, and the physical structure of materials at the molecular level.

Organic Chemistry: The study of substances containing carbon. Carbon is one of the most abundant elements on Earth and forms over twenty million known compounds. Most chemicals in all living organisms are based on carbon.

Inorganic Chemistry: The study of substances that are not primarily based on carbon. Inorganic chemicals are found in rocks and minerals. A current important area is the design of materials for energy and information technology.

Analytical Chemistry: The study of the composition of matter. It focuses on separating, identifying, and quantifying chemicals in samples. Analytical chemists use complex instruments to analyse unknown materials.

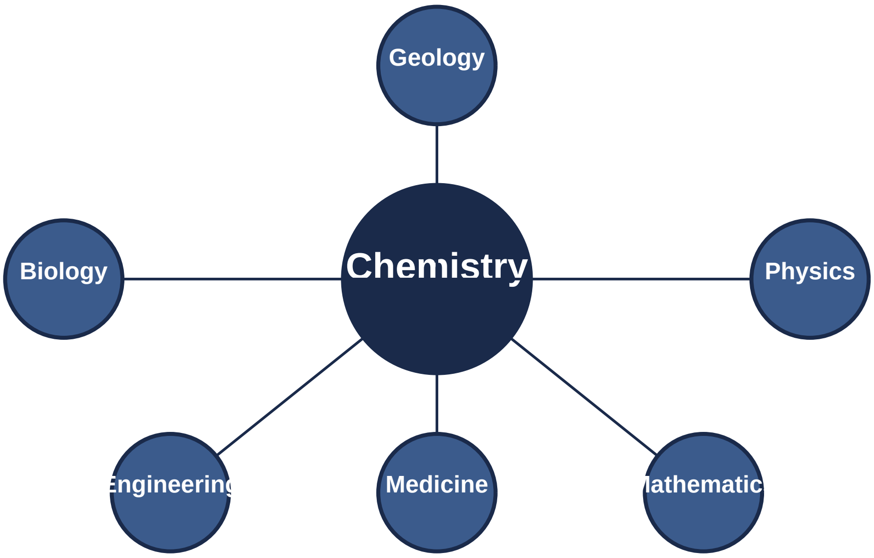
Biochemistry: The study of chemical processes that occur in living things. It covers everything from basic cellular processes to understanding disease states so that better treatments can be developed.

All of these disciplines involve taking measurements, making observations, and drawing conclusions. Because living and non■living things are made of matter, chemistry affects all aspects of life and most natural events.

The scope of chemistry extends to **agriculture**, **medicine**, **food production**, and **building construction**. Chemistry provides useful substances but can also produce dangerous ones, such as chlorofluorocarbons (CFCs), oxides of nitrogen, carbon, and sulphur, which negatively affect human life and the environment.

1.2 Relationship with Other Natural Sciences

Chemistry is one branch of science. Science is the process by which we learn about the natural universe through observation, testing, and generating models that explain our observations. Because the physical universe is vast, there are many branches of science – but there is significant overlap among them.



Key overlaps between chemistry and other sciences:

Biochemistry (Chemistry + Biology): The study of chemical processes occurring in living matter. Chemists work on isolation, characterisation, and biological activities of compounds from medicinal plants.

Geochemistry (Chemistry + Geology): The study of processes that control the abundance, composition, and distribution of chemical compounds and isotopes in geologic environments.

Physical Chemistry / Chemical Physics (Chemistry + Physics): The application of techniques and theories of physics to the study of chemical systems. It investigates physicochemical phenomena using atomic and molecular physics.

Medicinal Chemistry (Chemistry + Medicine): The design, development, and synthesis of pharmaceutical drugs.

Because all of these fields involve 'stuff' – matter – chemistry has been called the **central science**, linking them all together.

1.3 The Role Chemistry Plays in Production and Society

A. Agriculture

Chemistry has brought the world chemical fertilizers such as **calcium superphosphate**, **urea**, **ammonium sulphate**, and **sodium nitrate**. These chemicals have greatly increased the yield of fruits, vegetables, and other crops.

Chemistry has been effective in the manufacture of **pesticides** (fungicides, herbicides, and insecticides), which have lessened crop damage and helped supply the ever■growing demand for food. Chemistry also plays a role in manufacturing better■quality plastic pipes for irrigation, which has improved farming efficiency.

B. Food Production

Chemistry has led to the discovery of different kinds of **food preservatives** that greatly assist in preserving food products for longer periods. It has given methods to test for adulterants, ensuring the supply of pure foodstuffs.

Consumers have benefited from new technologies that have increased food availability, appearance, nutritional content, and flavour. A local example is the preservation of raw meat using salt and other chemicals to extend its shelf life.

C. Medicine

Chemistry has provided mankind with a large number of life-saving medicines. The discovery of **sulphur drugs** and **penicillin** provided cures for dysentery and pneumonia. Life-saving drugs like **cisplatin** and **Taxol** are effective for cancer therapy, and **AZT** is used to prolong the lives of AIDS victims.

Various classes of medicinal chemicals include:

- **Disinfectants** – kill microbes in toilets, floors, and drains. Sanitizers used during COVID-19 belong to this group.
- **Analgesics** (painkillers) – drugs used to achieve pain relief.
- **Anesthetics** – make medical operations more effective by relieving pain.
- **Antibiotics** – control infections and cure diseases.
- **Antiseptics** – prevent contamination of wounds by bacteria.
- **Tranquillizers** – reduce tension and bring calm to patients suffering from mental diseases.

D. Building Construction

By providing building resources such as **glass**, **steel**, and **cement**, chemistry helps in the construction of safer houses, multi-storey structures, and long-lasting dams and bridges.

A remarkable example is the **Grand Ethiopian Renaissance Dam (GERD)**, which is under construction in the Benishangul-Gumuz Region. The GERD is a 6,450 MW hydroelectric project – the largest in Africa – and it relies on chemistry for the production of concrete, steel, and other construction materials.

1.4 Some Common Chemical Industries in Ethiopia

An **industry** is an economic activity concerned with the processing of raw materials and the manufacture of goods in factories. The Ethiopian government has been heavily engaged in expanding industries over the past two decades, with several industrial parks under construction.

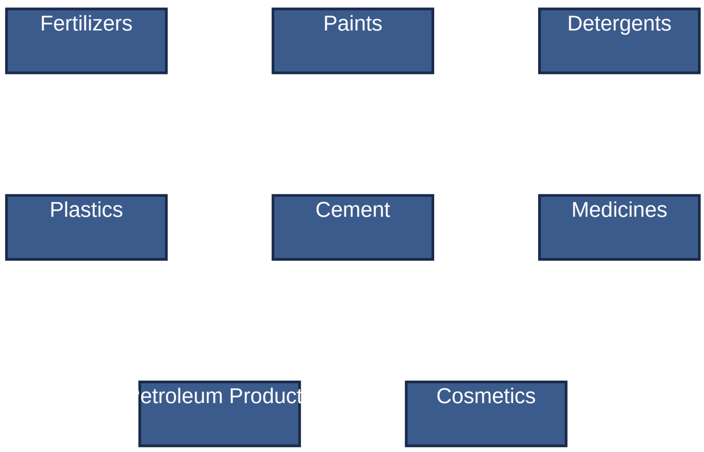
The chemical industries comprise companies that manufacture inorganic and organic industrial chemicals, explosives, fragrances, agrochemicals, polymers, and plastics.

No.	Enterprise Name	Location	Products
1	Chorra Gas & Chemical Products	A.A.	Plastics, chemicals, petroleum products
2	Chorra Gas & Chemical Products	A.A.	Aluminium sulphate and sulphuric acid
3	Ziway Caustic Soda Factory	Ziway	Sodium hydroxide
4	Abijata Soda Ash Factory	Bulbula	Trona ($Na_3H(CO_3)_2 \cdot 2H_2O$)
5	Repi Soap & Detergent P.L.C	A.A.	Soap and detergent

6	Adola Magnesium Oxide Factory	Adolla	Magnesium oxide
7	Adami Tulu Pesticide Processing Plant	Adami Tulu	Formulates malathion, endosulfan, diazinon, fenitrothion, dimethoate
8	Nefas Silk Paints Factory	A.A.	Paints, varnishes, antirusts, glues
9	Modern Building Industries	A.A.	Cement, ceramics, paints, sanitary ware, adhesives, plastics, rubber, terrazzo tiles
10	Kadisco Chemical Industry	A.A.	Paints, coatings, adhesives
11	Tadesse Filatea PLC	Woliso	Soap, detergent, corrugated iron, nails, infant milk formula
12	Etab Laundry Soap Factory	Hawassa	Soap and detergent
13	Get-Eshet Detergent and Leather Manufacturing	Bishoftu	Detergent products and chemical inputs
14	Ethio-Asia Industries S.C	A.A.	Soap and detergent
15	Y.B Cosmetics	Sheger City	Cosmetics and perfume
16	Mekab PLC (Cosmetics)	A.A.	Hair oil, shampoo, conditioner, body oil, vaseline, body lotion, detergents, plastics
17	BEKAS Chemicals PLC	A.A.	Detergents, cosmetic products, plastic packing materials, industrial surfactants, putty
18	Arbaminch Textile Share Company	Arbaminch	Textile and fabric products
19	Teamco Soap Factory	Burayu	Soap and detergent

Other chemical product industries in Ethiopia:

- **Cement:** Mugher, Diredawa, Mesobo, Derba, Midroc, Dangote
- **Sugar:** Metehara, Wonji, Finchaa, Omokuraz
- **Paper and Pulp:** Wonji
- **Pharmaceuticals:** Addis, Ethiopia, Adigrat
- **Tyre:** Horizon Addis Tyre



Unit Summary

- **Chemistry** is the science that deals with the properties, composition, and structure of substances, the transformations they undergo, and the energy changes involved.
- A **substance** is matter with uniform properties. Every substance has its own unique **properties**, **composition**, and **structure**.
- **Transformations** are changes in form, nature, or appearance that are accompanied by energy changes.

• The five main disciplines of chemistry are: **physical**, **organic**, **inorganic**, **analytical**, and **biochemistry**.

• Chemistry is the **central science** because it overlaps with and supports biology, physics, geology, medicine, and other fields.

• Chemistry plays a vital role in **agriculture** (fertilizers, pesticides), **food production** (preservatives, testing), **medicine** (drugs, disinfectants), and **building construction** (cement, steel, glass).

• Ethiopia has many chemical industries producing fertilisers, paints, detergents, plastics, cement, pharmaceuticals, cosmetics, and textiles.

• The **Grand Ethiopian Renaissance Dam (GERD)** is an example of large■scale construction made possible by chemistry.

• Chemistry provides essential substances for meeting basic human needs: food, clothing, shelter, health, energy, clean air, water, and soil.

• Chemistry can also produce dangerous substances that harm the environment (e.g., CFCs, oxides of nitrogen, carbon, and sulphur).

Key Terms

Analgesics – drugs used to relieve pain.
Antibiotics – drugs used to control infections.
Anesthetics – drugs that relieve pain during medical procedures.
Antiseptics – substances applied to wounds to prevent infection.
Chemical industry – companies that manufacture chemicals and chemical products.
Chemical products – substances produced by chemical processes.
Composition – the nature of something's ingredients or constituents.
Disinfectants – chemicals that kill microbes on surfaces.
Industry – economic activity concerned with processing raw materials.
Insecticides – pesticides specifically targeting insects.
Matter – physical substance that occupies space and has mass.
Property – an attribute, quality, or characteristic of a substance.
Structure – the arrangement of parts or elements of something complex.
Substance – a particular kind of matter with uniform properties.
Tranquillizers – drugs that reduce tension and anxiety.

End of Unit 1 – Chemistry and Its Importance